TRƯỜNG ĐẠI HỌC
VĂN LANG
Đạo đức - Ý chí - Sáng tạo

KHOA CÔNG NGHỆ THÔNG TIN

CƠ SỞ LẬP TRÌNH
(Programming Basics)

CHAPTER 5: FLOW CONTROL

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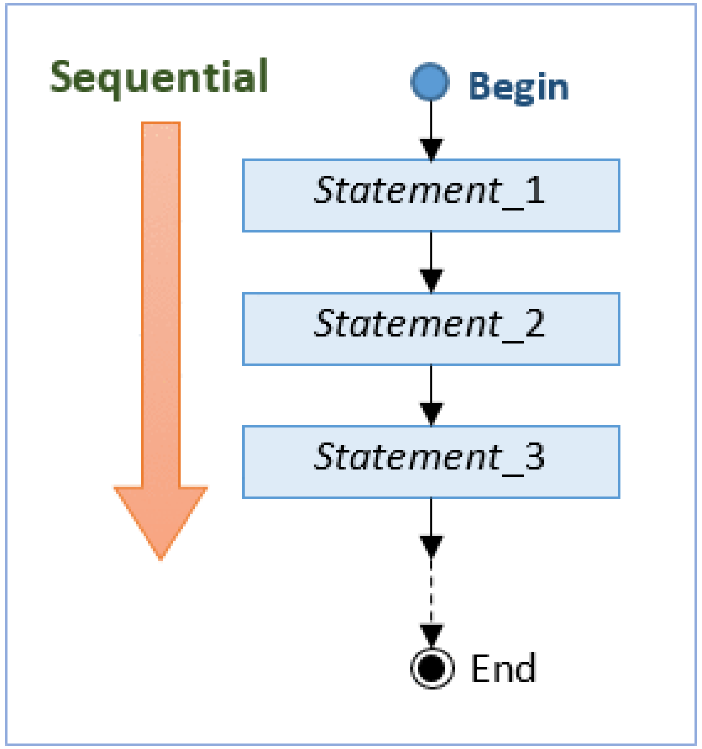
HỌC KỲ I

KHÓA:

Sequential Flow Control

Sequential flow is the most common and straight-forward, where programming statements are executed in the order that they are written - from top to bottom in a sequential manner.



Conditional Flow Control

Syntax	Example

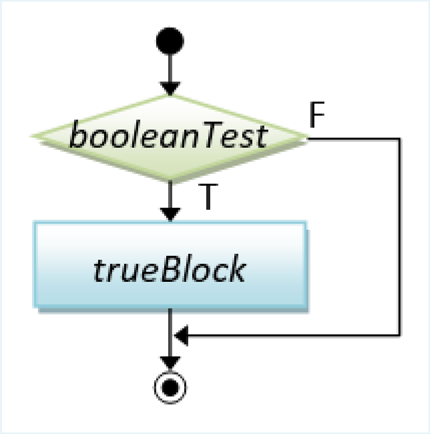
`// if-then`
`if (booleanTest) {`
`   trueBlock;`
`}`
`// next statement`
	`int mark = 80;`
	`if (mark >= 80) {`
	`   System.out.println("Well Done!");`
	`   System.out.println("Keep it up!");`
	`}`
	`System.out.println("Life goes on!);`
	
	`double temperature = 80.1;`
	`if (temperature > 80) {`
	`   System.out.println("Too Hot!");`
	`}`
	`System.out.println("yummy!");`

```java
// if-then-else
if (booleanTest) {
   trueBlock;
} else {
   falseBlock;
}
// next statement
```

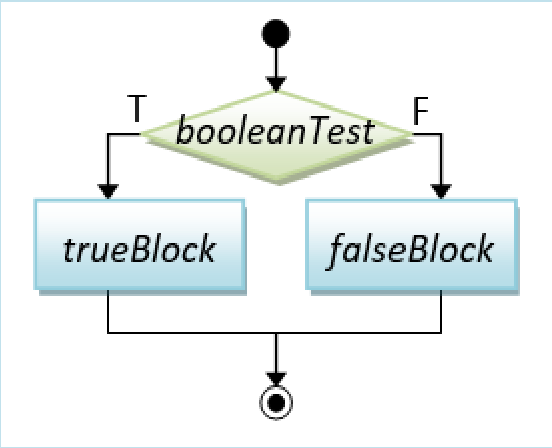
```java
int mark = 50;     // Assume that mark is [0, 100]
if (mark >= 50) {  // [50, 100]
   System.out.println("Congratulation!");
   System.out.println("Keep it up!");
} else {           // [0, 49]
   System.out.println("Try Harder!");
}
System.out.println("Life goes on!");

double temperature = 80.1;
if (temperature > 80) {
   System.out.println("Too Hot!");
} else {
   System.out.println("Too Cold!");
}
System.out.println("yummy!");
```

Flowchart



(1)



(2)

Braces:

You could omit the braces { }, if there is only one statement inside the block.

```
// if-then
int absValue = -5;
if (absValue < 0) absValue = -absValue;   // Only one statement in the block, can omit { }

int min = 0, value = -5;
if (value < min) {      // More than one statements in the block, need { }
   min = value;
   System.out.println("Found new min");
}

// if-then-else
int mark = 50;
if (mark >= 50)
   System.out.println("PASS");   // Only one statement in the block, can omit { }
else {                           // More than one statements in the block, need { }
   System.out.println("FAIL");
   System.out.println("Try Harder!");
}

// Harder to read without the braces
int number1 = 8, number2 = 9, absDiff;
if (number1 > number2) absDiff = number1 - number2;
else absDiff = number2 - number1;
```

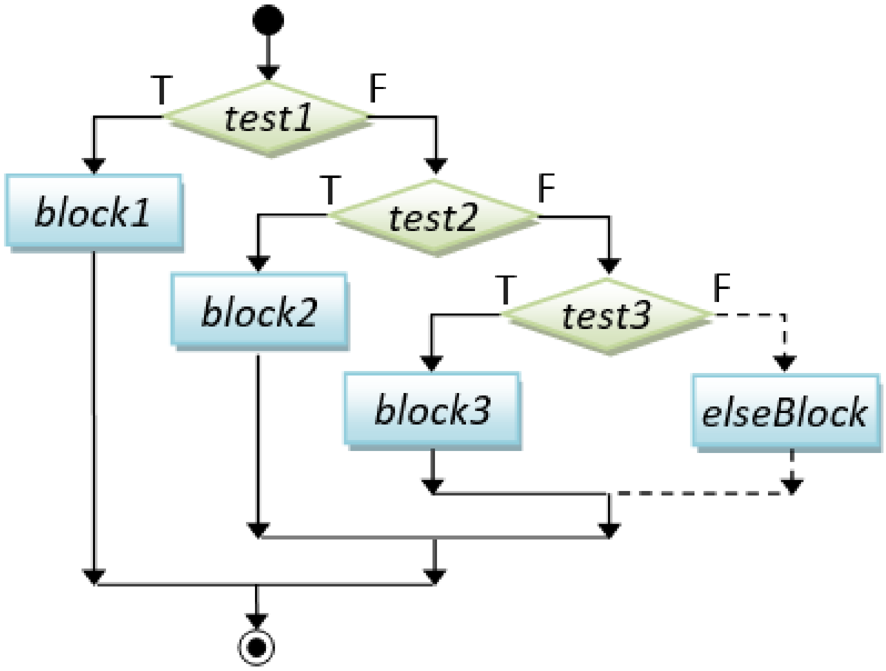
I recommend that you keep the braces to improve the readability of your program, even if there is only one statement in the block.

Nested-if

Syntax	Example

`// nested-if`
`if (booleanTest1) {`
`   block1;`
`} else if (booleanTest2) {`
`   block2;`
`} else if (booleanTest3) {`
`   block3;`
`} else if (booleanTest4) {`
`   ......`
`} else {`
`   elseBlock;`
`}`
`// next statement`
	`int mark = 62;  // Assume that mark is [0, 100]`
	`if (mark >= 80) {          // [80, 100]`
	`   System.out.println("A");`
	`} else if (mark >= 65) {    // [65, 79]`
	`   System.out.println("B");`
	`} else if (mark >= 50) {    // [50, 64]`
	`   System.out.println("C");`
	`} else {                    // [0, 49]`
	`   System.out.println("F");`
	`}`
	`System.out.println("Life goes on!");`
	
	`double temperature = 61;`
	`if (temperature > 80) {          // > 80`
	`   System.out.println("Too Hot!");`
	`} else if (temperature > 75) { // (75, 80]`
	`   System.out.println("Just right!");`
	`} else {                         // <= 75`
	`   System.out.println("Too Cold!");`
	`}`
	`System.out.println("yummy!");`

Flowchart



Run and explain code

(1)

```
int i = 0, j = 0;
if (i == 0)       // outer-if
   if (j == 0)    // inner-if
      System.out.println("i and j are zero");
else System.out.println("xxx");   // This else can pair with the inner-if and outer-if?!
```

(2)

```
int i = 0, j = 0;
if (i == 0)
   if (j == 0)
      System.out.println("i and j are zero");
   else    // associated with if (j == 0) - the nearest if
      System.out.println("xxx");
```

(3)

```
// Force the else-clause to associate with the outer-if with parentheses
int i = 0, j = 0;
if (i == 0) {
   if (j == 0)
      System.out.println("i and j are zero");
} else {
   System.out.println("i is not zero");   // non-ambiguous for outer-if
}
```

(4)

```
// Force the else-clause to associate with the inner-if with parentheses
int i = 0, j = 0;
if (i == 0) {
   if (j == 0) {
      System.out.println("i and j are zero");
   } else {
      System.out.println("i is zero, j is not zero");   // non-ambiguous for inner-if
   }
}
```

Nested-if vs. Sequential-if

```
// Assume mark is between 0 and 100
// This "sequential-if" works but NOT efficient!
// Try mark = 81, which will run thru ALL the if's.
int mark = 81;
if (mark > 80) {  // [81, 100]
   grade = 'A';
}
if (mark > 65 && mark <= 80) {  // [66, 80]
   grade = 'B';
}
if (mark >= 50 && mark <= 65) { // [50, 65]
   grade = 'C';
}
if (mark < 50) {  // [0, 49]
   grade = 'F';
}
```

```
// This "nested-if" is BETTER
// Try mark = 81, which only run thru only the first if.
int mark = 81;
if (mark > 80) {   // [81, 100]
   grade = 'A';
} else if (mark > 65 && mark <= 80) {  // [66, 80]
   grade = 'B';
} else if (mark >= 50 && mark <= 65) { // [50, 65]
   grade = 'C';
} else {
   grade = 'F';    // [0, 49]
}
```

```
// This "nested-if" is the BEST with fewer tests
int mark = 81;
if (mark > 80) {          // [81, 100]
   grade = 'A';
} else if (mark > 65) {  // [66, 80]
   grade = 'B';
} else if (mark >= 50) { // [50, 65]
   grade = 'C';
} else {                  // [0, 49]
   grade = 'F';
}
```

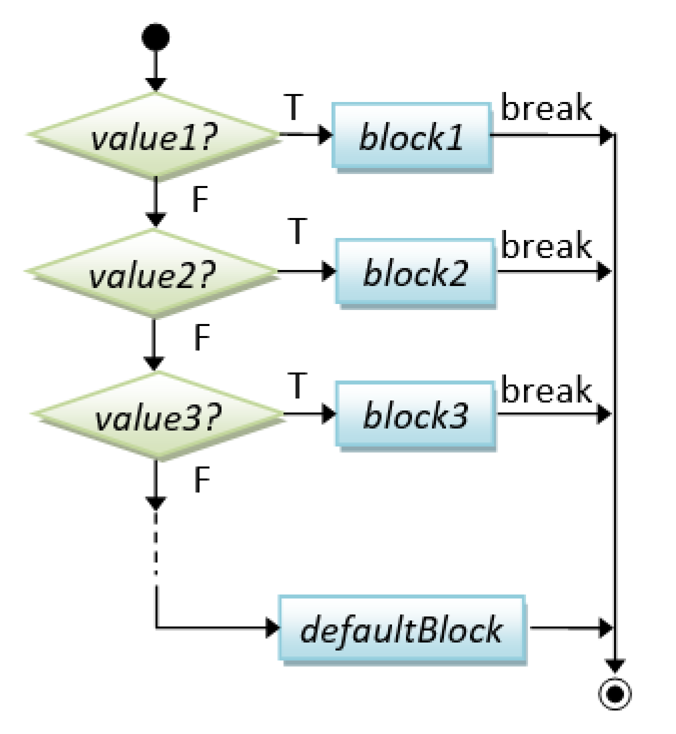
(*)

switch-case-default

Syntax

```
// switch-case-default
switch (selector) {
   case value1:
      block1; break;
   case value2:
      block2; break;
   case value3:
      block3; break;
   ......
   case valueN:
      blockN; break;
   default: // not the above
      defaultBlock;
}
```

Flowchart



Example

```java
// Print number in word
int number = 3;
switch (number) {   // "int" selector
   case 1:   // "int" value
      System.out.println("ONE"); break;
   case 2:
      System.out.println("TWO"); break;
   case 3:
      System.out.println("THREE"); break;
   default:
      System.err.println("OTHER");
}
```

(1)

```java
// Select arithmetic operation
char operator = '*';
int num1 = 5, num2 = 8, result;
switch (operator) {   // "char" selector
   case '+':   // "char" value
      result = num1 + num2; break;
   case '-':
      result = num1 - num2; break;
   case '*':
      result = num1 * num2; break;
   case '/':
      result = num1 / num2; break;
   default:
      System.out.println("Unknown operator);
}
```

(2)

Conditional Expression (... ? ... : ...)

Syntax

```
// Conditional Expression
booleanExpr ? trueExpr : falseExpr
  // An expression that returns
  // the value of trueExpr
  // or falseExpr
```

Conditional expression is a short-hand for if-else. But you should use it only for one-liner, for readability.

Examples

```
int num1 = 9, num2 = 8, max;
max = (num1 > num2) ? num1 : num2;  // RHS returns num1 or num2
// same as
if (num1 > num2) {
   max = num1;
} else {
   max = num2;
}

int value = -9, absValue;
absValue = (value > 0) ? value : -value;  // RHS returns value or -value
// same as
if (value > 0) absValue = value;
else absValue = -value;

int mark = 48;
System.out.println((mark >= 50) ? "PASS" : "FAIL");  // Return "PASS" or "FAIL"
// same as
if (mark >= 50) System.out.println("PASS");
else System.out.println("FAIL");
```

Exercise

1. Check the smallest number between 2 numbers.
2. Check the largest number between 3 numbers
3. Print out whether a number is even or odd

   Hint: numbers divisible by 2 are even numbers, numbers that are divisible by 2 are odd numbers
4. Enter the number of points, and indicate whether the score is Good, Good, Average, Weak, know that:

   Good: [9 – 10]; Fair: [7 – 9); Average: [5 – 7); Weak: [0 – 5). Otherwise, show message "Invalid score"
5. Create an English dictionary. When the user enters an English word to look up, the Vietnamese meaning of that word will appear on the screen, with an example attached.

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